Homework 3 – Regression

1 Question 1

Below, I simulate a rather quite annoying set of regression data.

```
set.seed(0510)
n <- 250
dat.q1 <- tibble(x1 = rnorm(n, mean = 50, sd = 10),
                 x2 = factor(sample(c("A", "B"), n, replace = TRUE)),
                 x3 = factor(sample(c("Control", "Treat"), n, replace = TRUE)),
                 x4 = factor(sample(c("Low", "High"), n, replace = TRUE), levels=c("Low", "High")),
                 x5 = rnorm(n, mean = 0, sd = 1),
                 x6 = factor(sample(c("Group1", "Group2"), n, replace = TRUE)),
                 x7 = rnorm(n, mean = 10, sd = 2),
                 x8 = rnorm(n, mean = 10, sd = 2)) |>
        mutate(y = 5 +                          # Intercept
                   # Quantiative x1
                   2 * x1 +
                   # Categorical x2
                   (x2 == "B") * 10 +
                   # Categorical x Categorical (x3*x4)
                   (x3 == "Treat") * 5 + (x4 == "High") * 2 +
                   (x3 == "Treat" & x4 == "High") * 15 +
                   # Quantitative x Categorical  (x5*x6)
                   3 * x5 + (x6 == "Group2") * 4 +
                   (x5 * (x6 == "Group2")) * 8 +
                   # Quantitative x Quantitative (x7*x8)
                   1.5 * x7 + 0.5 * x8 +
                   0.8 * (x7 * x8) +
                   # Error
                   rnorm(n, mean = 0, sd = 15))
```

1.1 Part a

Fit the regression model.

$$
\begin{aligned}
Y_i = &\beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \beta_3 I(x_{3i} = \text{Treat}) + \beta_4 I(x_{4i} = \text{High}) + \\
&\beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\
&\beta_9 (I(x_{3i} = \text{Treat}) \times I(x_{4i} = \text{High})) + && \text{(Categorical} \times \text{Categorical)} \\
&\beta_{10} (x_{5i} \times I(x_{6i} = \text{Group2})) + && \text{(Continuous} \times \text{Categorical)} \\
&\beta_{11} (x_{7i} \times x_{8i}) + \epsilon_i && \text{(Continuous} \times \text{Continuous)}
\end{aligned}
$$

Solution: First, we fit the regression model using `lm()`.

```
mod.q1 <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.q1)
```

1.2 Part b

Conduct a residual and outlier analysis.

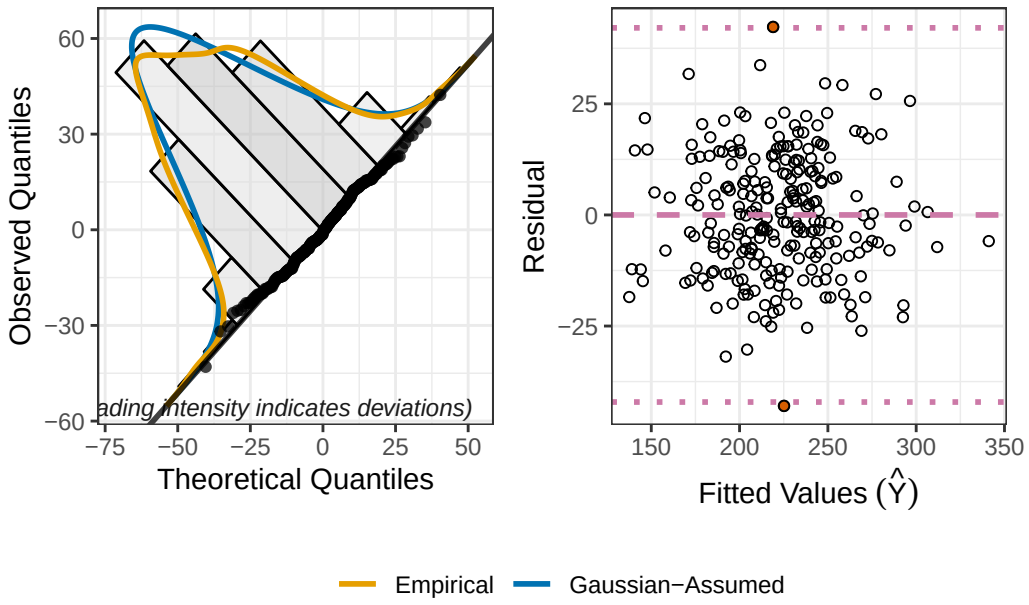
Solution The first 2 graphs show the residual analysis for the regression model. The Q-Q plot follows the expected pattern pretty well in the middle, close to the gaussian curve, but deviated a little in the middle. The residual versus fitted plot shows that the residuals are scattered around zero without a clear nonliniear patter. It is reasonable to assume that variance is approximately constant.

The next 4 graphs show outlier analysis. Although we can see several points have relatively high leverage, the Cook's distance values show us that they are below the threshold of suggesting strong influence. The DFFITS plot identifies

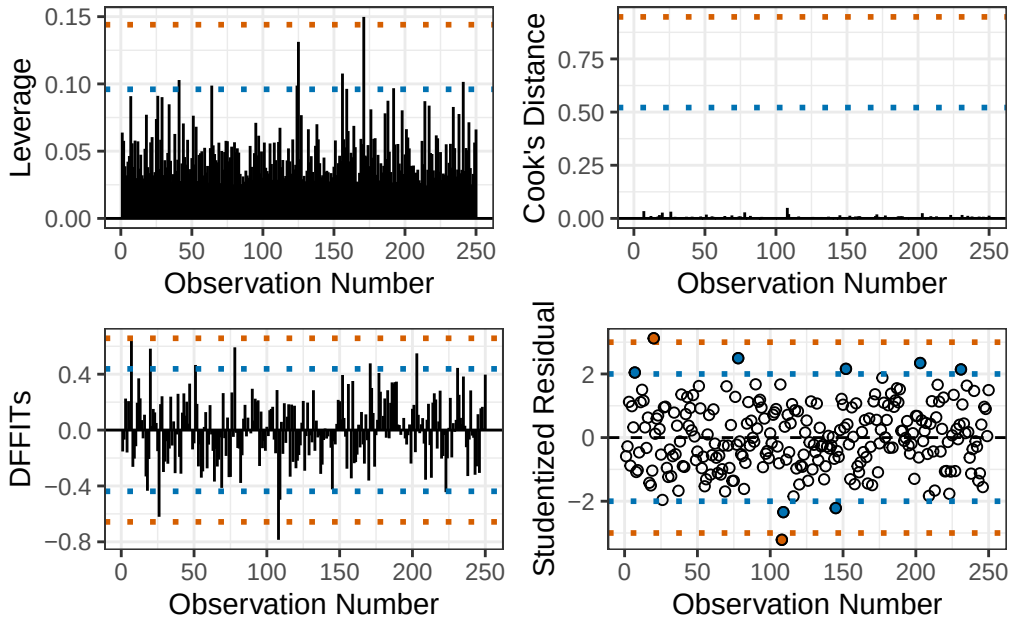
a few observations that exceed the reference thresholds, indicating potential influence on their fitted values and the studentized residual plot shows a number of potential outliers, including 2 observations with residuals beyond just a moderate outlier.

```
source("plotResiduals.R")
source("plotInfluence.R")

plotResiduals(mod.q1)
```




```
plotInfluence(mod.q1)
```



1.3 Part c

Interpret the intercept, the coefficients on x_1 and x_2.

Solution intercept: The estimated intercept is 8.89. This is the predicted value of y when all other predictors are equal to zero and the categorical predictors are at their reference levels. coefficient on x_1: The estimated coefficient is 2.1. Holding all other variables constant, a 1 unit increase in x_1 leads to an estimated increase of 2.1 units in y. coefficient on x_2: The estimated coefficient is 10.02 with category A as the reference level. Holding all other variables constant, observations in category B and predicted to have y values approximately 10.02 units higher than observations in category A.

```
summary(mod.q1)
```

```
Call:
lm(formula = y ~ x1 + x2 + x3 * x4 + x5 * x6 + x7 * x8, data = dat.q1)

Residuals:
    Min      1Q  Median      3Q     Max
-42.979  -9.395  -1.054  10.150  42.305

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     8.89653   26.94440   0.330 0.741554
x1              2.13546    0.08944  23.876  < 2e-16 ***
x2B            10.02851    1.79959   5.573 6.77e-08 ***
x3Treat         6.51162    2.66290   2.445 0.015198 *
x4High          3.52182    2.53119   1.391 0.165413
x5              2.48350    1.29418   1.919 0.056183 .
x6Group2        4.98495    1.80989   2.754 0.006336 **
x7              0.37174    2.61136   0.142 0.886920
x8             -0.92790    2.60484  -0.356 0.721990
x3Treat:x4High 15.19194    3.66722   4.143 4.77e-05 ***
x5:x6Group2     6.94743    1.85460   3.746 0.000225 ***
x7:x8           0.92871    0.25937   3.581 0.000415 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.03 on 238 degrees of freedom
Multiple R-squared:  0.8562,    Adjusted R-squared:  0.8495
F-statistic: 128.8 on 11 and 238 DF,  p-value: < 2.2e-16
```

1.4 Part d

Untangle and interpret the $x_3 * x_4$ interaction using algebra. Use Table 1 and Figure 1 for context.

Solution

$$Y_i = \beta_3 I(x_{3i} = \text{Treat}) + \beta_4 I(x_{4i} = \text{High}) + \beta_9 (I(x_{3i} = \text{Treat}) \times I(x_{4i} = \text{High})) + C$$

where C is everything else in the equation.

$E(Y|\text{control, low}) = C$
$E(Y|\text{control, high}) = \beta_4 + C$
$E(Y|\text{treat, low}) = \beta_3 + C$
$E(Y|\text{treat, high}) = \beta_3 + \beta_4 + \beta_9 + C$

β_3: Compare Treat and Control when the condition is Low. The difference is β_3.

β_4: Compare High and Low when the group is Control. The difference is β_4.

β_9: The interaction tells us how much the treatment effect changes between Low and High.

The interaction between x_3 and x_4 indicates that the effect of treatment on y depends on whether the condition is Low or High. When the condition is Low, the estimated treatment effect is $\hat{\beta}_3 = 6.5116$. When the condition is

High, the estimated treatment effect is $\hat{\beta}_3 + \hat{\beta}_9 = 6.5116 + 15.1919 = 21.7035$. Therefore, the interaction coefficient $\hat{\beta}_9 = 15.1919$ represents the difference between these two treatment effects.

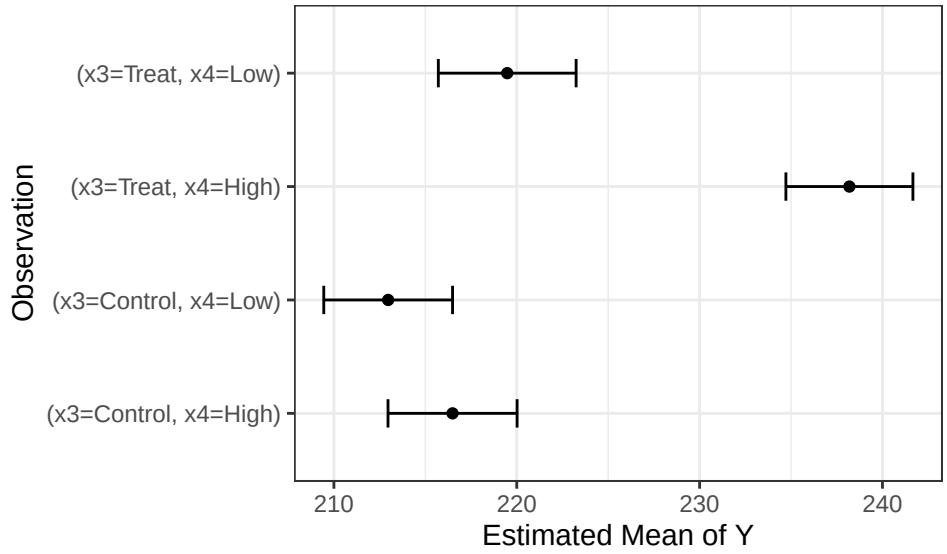
```r
library(emmeans)
emm.q1d <- emmeans(mod.q1, ~x3*x4) |> # Don't average across levels: use |x2*x6
            data.frame() |>
            rename("Mean"  = "emmean",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL")
```

Table 1: Estimated marginal means for the model in Question 1.

x3	x4	Mean	SE	df	Lower	Upper
Control	Low	212.9676	1.7871	238	209.4470	216.4882
Treat	Low	219.4792	1.9123	238	215.7120	223.2464
Control	High	216.4894	1.7916	238	212.9601	220.0188
Treat	High	238.1930	1.7631	238	234.7198	241.6662

```r
emm.q1d <- emm.q1d |>
  mutate(Observation = paste("(x3=", x3, ", ", "x4=", x4, ")", sep=""))
ggplot(emm.q1d) +
  geom_point(aes(x=Mean, y=Observation))+
  geom_errorbar(aes(y=Observation,
                    xmin=Lower, xmax=Upper),
                width = 0.25)+
  # facet_wrap(~x2*x6)+
  theme_bw()+
  xlab("Estimated Mean of Y")
```

Figure 1: Estimated marginal means for the model in Question 1.



1.5 Part e

Untangle and interpret the $x_5 * x_6$ interaction using algebra. Use Table 2 and Figure 2 for context.

Solution

$$Y_i = \beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_{10}(x_{5i} \times I(x_{6i} = \text{Group2})) + C$$

where C is everything else in the equation.

For Group1, $(I(x_6 = \text{Group2}) = 0)$, so

$E(Y \mid x_6 = \text{Group1}) = \beta_5 x_5 + C.$

The slope of x_5 for Group1 is therefore β_5.

For Group2, $(I(x_6 = \text{Group2}) = 1)$, so

$E(Y \mid x_6 = \text{Group2}) = \beta_5 x_5 + \beta_6 + \beta_{10} x_5 + C \ = (\beta_5 + \beta_{10}) x_5 + \beta_6 + C.$

The slope of x_5 for Group 2 is therefore $\beta_5 + \beta_{10}$. The interaction coefficient β_{10} represents the difference between the two slopes. Using Table 2, this difference is $9.4309 - 2.4835 = 6.9474$. Thus, the positive interaction indicates that the relationship between x_5 and y is approximately 6.95 units stronger for Group 2 than for Group 1. Figure 2 supports this interpretation because the Group 2 line is steeper than the Group 1 line.

```
emm.q1e <- emtrends(mod.q1, var="x5", ~x5|x6) |>
            data.frame() |>
            rename("Slope"  = "x5.trend",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL") |>
            dplyr::select(x6, Slope, SE, df, Lower, Upper)
```

Table 2: Estimated marginal trends for the model in Question 1.

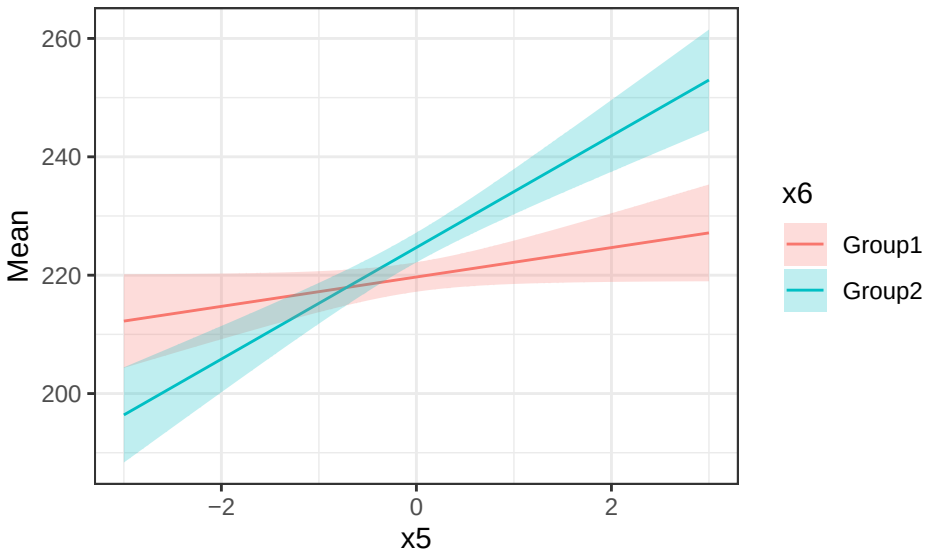
x6	Slope	SE	df	Lower	Upper
Group1	2.4835	1.2942	238	-0.0660	5.0330
Group2	9.4309	1.3367	238	6.7977	12.0642

In Figure 2, the slopes are the same and just the intercepts vary depending on x_2, x_3, and x_4.

```
emm.q1e <- emmeans(mod.q1, ~x5*x6,
                   at = list(x5 = seq(-3, 3, length.out=500))) |> # Don't average across levels: use |x2*x3*x4
  data.frame() |>
            rename("Mean"  = "emmean",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL")

ggplot(emm.q1e) +
  geom_line(aes(x=x5, y=Mean, color=x6))+
  geom_ribbon(aes(x=x5, ymin=Lower, ymax=Upper, fill=x6), alpha=0.25)+
  # facet_wrap(~x2*x3*x4)+
  theme_bw()
```

Figure 2: Estimated marginal trends for the model in Question 1.



1.6 Part f

Untangle and interpret the $x_7 * x_8$ interaction using algebra. Use Table 3 and Table 4 for context.

Solution

$$Y_i = \beta_7 x_{7i} + \beta_8 x_{8i} + \beta_{11}(x_{7i} \times x_{8i}) + C$$

Grouping the terms involving x_7, we get

$Y = x_7(\beta_7 + \beta_{11} x_8) + \beta_8 x_8 + C.$

Therefore, the effect of x_7 depends on x_8, through the coefficient $\beta_7 + \beta_{11} x_8$.

Similarly, grouping the terms involving x_8, we get

$Y = x_8(\beta_8 + \beta_{11} x_7) + \beta_7 x_7 + C.$

Therefore, the effect of x_8 depends on x_7, through the coefficient $\beta_8 + \beta_{11} x_7$.

Tables 3 and 4 support this interpretation. In Table 3, the estimated slope of x_7 increases from 7.9230 to 11.5027 as x_8 increases. In Table 4, the estimated slope of x_8 increases from 6.8569 to 10.2716 as x_7 increases. These results indicate a positive interaction: as either variable increases, the effect of the other variable on Y becomes stronger.

```
emm.q1f.x7 <- emtrends(mod.q1, var="x7", ~x7|x8,
                       at = list(x8 = mean(dat.q1$x8) + c(-1, 0 ,1)*sd(dat.q1$x8))) |>
              data.frame() |>
              rename("Slope"  = "x7.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 3: Estimated marginal trends for the model in Question 1.

x8	Slope	SE	df	Lower	Upper
8.1309	7.9230	0.6751	238	6.5932	9.2529
10.0582	9.7129	0.5015	238	8.7250	10.7008
11.9854	11.5027	0.7396	238	10.0458	12.9597

```
emm.q1f.x8 <- emtrends(mod.q1, var="x8", ~x8|x7,
                       at = list(x7 = mean(dat.q1$x7) + c(-1, 0 ,1)*sd(dat.q1$x7))) |>
              data.frame() |>
              rename("Slope"  = "x8.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 4: Estimated marginal trends for the model in Question 1.

x7	Slope	SE	df	Lower	Upper
8.3824	6.8569	0.6069	238	5.6613	8.0525
10.2208	8.5643	0.4744	238	7.6297	9.4989
12.0592	10.2716	0.7325	238	8.8286	11.7146

A Johnson Neyman analysis using the `interactions` package for `R` (Long 2024), shows that the slope of x_7 is statistically discernible for all observed values of x_8 (Figure 3), and vice versa (Figure 4).

```
library(interactions)
johnson_neyman(mod.q1, pred = x7, modx = x8, control.fdr = T)
```

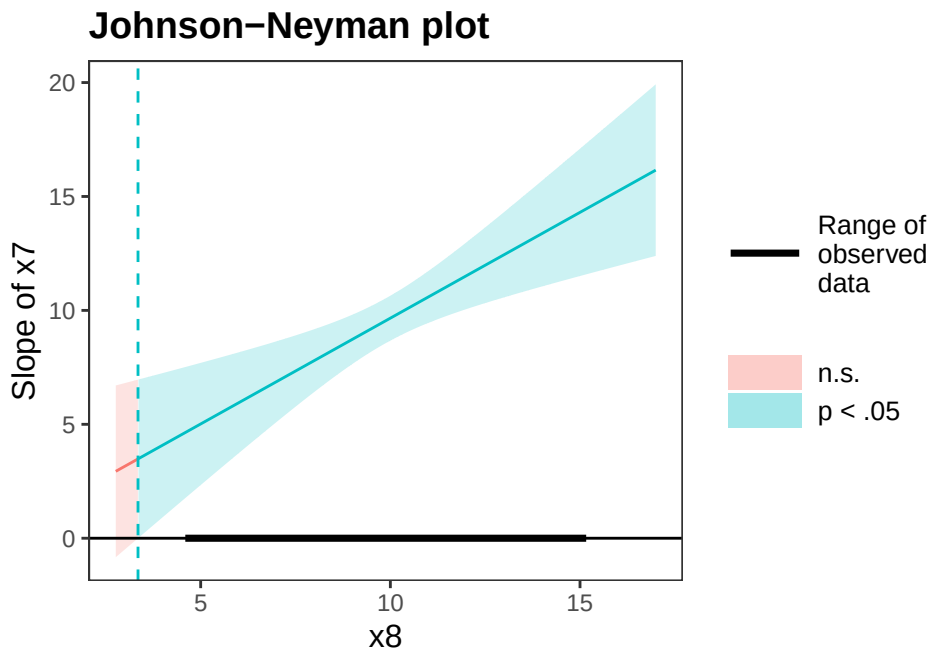
```
JOHNSON-NEYMAN INTERVAL

When x8 is OUTSIDE the interval [-13.08, 3.35], the slope of x7 is p < .05.

Note: The range of observed values of x8 is [4.69, 15.07]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 3: Johnson Neyman interval for the model in Question 1.




```
johnson_neyman(mod.q1, pred = x8, modx = x7, control.fdr = T)
```

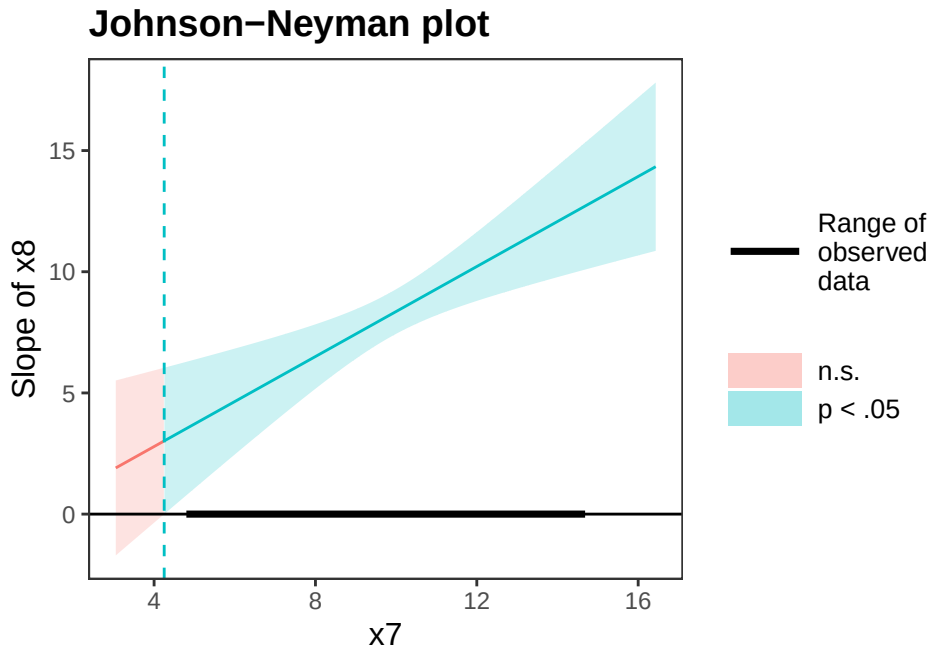
```
JOHNSON-NEYMAN INTERVAL

When x7 is OUTSIDE the interval [-9.96, 4.25], the slope of x8 is p < .05.

Note: The range of observed values of x7 is [4.89, 14.60]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 4: Johnson Neyman interval for the model in Question 1.



1.7 Part g

Standardize the continuous data and refit the regression model. Redo part f. What do you notice about the coefficient estimates and why does this make the untangling process easier?

Solution: First, we fit the regression model using `lm()`.

```
# Standardize Data
dat.z <- dat.q1 |>
  mutate(across(c(x1, x5, x7, x8), ~ as.vector(scale(.))))
# Refit Model
mod.q1z <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.z)
```

```
emm.q1g.z7 <- emtrends(mod.q1z, var="x7", ~x7|x8,
                    at = list(x8 = c(-1, 0 ,1))) |>
          data.frame() |>
          rename("Slope"  = "x7.trend",
                 "Lower" = "lower.CL",
                 "Upper" = "upper.CL") |>
          dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 5: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x8	Slope	SE	df	Lower	Upper
-1	14.57	1.24	238	12.12	17.01
0	17.86	0.92	238	16.04	19.67
1	21.15	1.36	238	18.47	23.83

```
emm.q1g.z8 <- emtrends(mod.q1z, var="x8", ~x8|x7,
                     at = list(x7 = c(-1, 0 ,1)))|>
              data.frame() |>
              rename("Slope"  = "x8.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 6: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x7	Slope	SE	df	Lower	Upper
-1	13.22	1.17	238	10.91	15.52
0	16.51	0.91	238	14.70	18.31
1	19.80	1.41	238	17.01	22.58

2 Question 2

Cui and Mao (2025) investigate how freezing and cryogenic temperatures affect the strength, flexibility, stiffness, and internal breaking behavior of 3D-printed CF/PEEK composites.

Carbon fiber reinforced PEEK (CF/PEEK) is a high-performance composite used in 3D printing for aerospace, deep-sea, polar research. The researchers aimed to assess whether and how extremely cold environments impact this composite to understand how 3D-printed CF/PEEK performs in freezing conditions. This work would be important to understand its limits and develop preventative methods to protect against structural failures.

The researchers created fifteen 3D printed dog-bone-shaped test specimens and placed them in a chamber cooled with liquid nitrogen to 25°C, −25°C, −75°C, −125°C, and −175°C. To test strength of the specimen, a machine pulled both ends of the specimen until it snapped.

In the table below, the researchers record the average tensile strength for three specimens from each temperature level.

Temperature	Mean Tensile Strength
25°C	81.93 MPa
−25°C	76.63 MPa

Temperature	Mean Tensile Strength
-75°C	63.23 MPa
-125°C	32.13 MPa
-175°C	12.83 MPa

2.1 Part a

Compute the estimates we derived using calculus in lecture 16:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)^2}.$$

2.2 Part b

Compute the estimates we derived using linear algebra in lecture 16:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

Hint: Run `?matrix()` in `R` for a quick reminder about how to create a matrix.

2.3 Part c

Compute the maximum likelihood estimates using `optim()` in `R` as described in lecture 16:

$$\max_{\beta_0,\beta_1,\sigma} \mathcal{L}(\theta = \{\beta_0, \beta_1, \sigma\}).$$

2.4 Part d

Fit the model using `lm()` and plot the resulting model. Compare to the results from parts (a) through (c).

3 Question 3

Bracht et al. (2016) evaluate MFAP4 as a biomarker for severity of fibrosis in hepatitic C patients.

```
1 dat.mfap4 <- read_csv("Bracht2016.csv")
```

3.1 Part a

Fit the model predicting `MFAP4` using `Fibrosis.Stage`. Write out the resulting estimate model and interpret the coefficients.

3.2 Part b

Conduct a residual analysis on the model in part a to demonstrate how to identify when you need a log transformation.

3.3 Part c

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

3.4 Part d

Conduct a residual analysis on the model in part c to demonstrate that the conditions for using regression are met.

3.5 Part e

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`, `Age`, and `Gender`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

4 Question 4

4.1 Part a

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Gaussian}(\mu = 0, \sigma = 1) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

4.2 Part b

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Shifted Exponential}(\lambda = 1, \mu = 0) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

Hint: Use the `tolerance::r2exp()` function with `rate=1` and `shift=-1`.

4.3 Part c

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Cauchy}(m = 0, s = 1) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

References

Bracht, Thilo, Christian Mölleken, Maike Ahrens, et al. 2016. "Evaluation of the Biomarker Candidate MFAP4 for Non-Invasive Assessment of Hepatic Fibrosis in Hepatitis C Patients." *Journal of Translational Medicine* 14 (1): 201.

Cui, Mengyang, and Jian Mao. 2025. "Effect of Low Temperature on Mechanical Properties and Fracture Behavior of 3D Printed Short Carbon Fiber Reinforced Polyether Ether Ketone Composites (CF/PEEK)." *Scientific Reports* 15 (1): 36244.

Long, Jacob A. 2024. *Interactions: Comprehensive, User-Friendly Toolkit for Probing Interactions*. https://doi.org/10.32614/CRAN.package.interactions.